

7.1 Verify Parameter Store Values

```
aws ssm get-parameters \  
--names /lab/db/endpoint /lab/db/port /lab/db/name \  
--with-decryption
```

```
{  
  "Parameters": [  
    {  
      "Name": "/lab/db/endpoint",  
      "Type": "String",  
      "Value": "lab-1-rds.cwuz3uwolwom.us-east-1.rds.amazonaws.com",  
      "Version": 1,  
      "LastModifiedDate": "2026-02-08T15:43:29.253000-05:00",  
      "ARN": "arn:aws:ssm:us-east-1:068259748700:parameter/lab/db/endpoint",  
      "DataType": "text"  
    },  
    {  
      "Name": "/lab/db/name",  
      "Type": "String",  
      "Value": "rds",  
      "Version": 1,  
      "LastModifiedDate": "2026-02-08T15:38:06.636000-05:00",  
      "ARN": "arn:aws:ssm:us-east-1:068259748700:parameter/lab/db/name",  
      "DataType": "text"  
    },  
    {  
      "Name": "/lab/db/port",  
      "Type": "String",  
      "Value": "3306",  
      "Version": 1,  
      "LastModifiedDate": "2026-02-08T15:43:29.251000-05:00",  
      "ARN": "arn:aws:ssm:us-east-1:068259748700:parameter/lab/db/port",  
      "DataType": "text"  
    }  
  ],  
  "InvalidParameters": []  
}
```

7.2 Verify Secrets Manager Value

```
aws secretsmanager get-secret-value \  
--secret-id lab-1/rds/mysql \  
--output json
```

```
mmauv@lonbrajdife MINGW64 ~/Documents/TheoWAF/class7/AWS/Armageddon/Class-7-Armageddon/lab-1/lab-1b (mike-armageddon-branch)  
$ aws secretsmanager get-secret-value --secret-id lab-1/rds/mysql --output json  
{  
  "ARN": "arn:aws:secretsmanager:us-east-1:068259748700:secret:lab-1/rds/mysql-5aDq8r",  
  "Name": "lab-1/rds/mysql",  
  "VersionId": "18e5d341-e161-49e0-9556-3a88b3708b49",  
  "SecretString": "{\"dbname\":\"rds\",\"engine\":\"mysql\",\"host\":\"lab-1-rds.cwuz3uwolwom.us-east-1.rds.amazonaws.com\",\"password\":\"TKOrpxKjGyu36joema\",\"port\":\"3306\",\"username\":\"admin\"}",  
  "VersionStages": [  
    "AWSCURRENT"  
  ],  
  "CreateDate": "2026-02-08T15:55:12.248000-05:00"  
}
```

7.3 Verify EC2 Can Read Both Systems

From EC2:

```
aws ssm get-parameter --name /lab/db/endpoint
```

```
{
  "Parameter": {
    "Name": "/lab/db/endpoint",
    "Type": "String",
    "Value": "lab-1-rds.cwuz3uwolwom.us-east-1.rds.amazonaws.com",
    "Version": 1,
    "LastModifiedDate": "2026-02-08T15:43:29.253000-05:00",
    "ARN": "arn:aws:ssm:us-east-1:068259748700:parameter/lab/db/endpoint",
    "DataType": "text"
  }
}
```

aws secretsmanager get-secret-value --secret-id lab-1/rds/mysql

```
PS C:\Users\mmauv> aws secretsmanager get-secret-value --secret-id lab-1/rds/mysql
{
  "ARN": "arn:aws:secretsmanager:us-east-1:068259748700:secret:lab-1/rds/mysql-5aDq8r",
  "Name": "lab-1/rds/mysql",
  "VersionId": "75aa1230-ac0f-4f8b-bb28-6fb3d4ec01ef",
  "SecretString": "{\"dbname\":\"rds\",\"engine\":\"mysql\",\"host\":\"lab-1-rds.cwuz3uwolwom.us-east-1.rds.amazonaws.com\",\"password\":\"TK0rpxKjGyu36956kJ\",\"port\":\"3306\",\"username\":\"admin\"}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2026-02-08T16:24:52.883000-05:00"
}
```

7.4 Verify CloudWatch Log Group Exists

aws logs describe-log-groups --log-group-name-prefix /aws/ec2/lab-1-rds-app

```
PS C:\Users\mmauv> aws logs describe-log-groups --log-group-name-prefix /aws/ec2/lab-1-rds-app
{
  "logGroups": [
    {
      "logGroupName": "/aws/ec2/lab-1-rds-app",
      "creationTime": 1770583086591,
      "retentionInDays": 1,
      "metricFilterCount": 1,
      "arn": "arn:aws:logs:us-east-1:068259748700:log-group:/aws/ec2/lab-1-rds-app:*",
      "storedBytes": 0,
      "logGroupClass": "STANDARD",
      "logGroupArn": "arn:aws:logs:us-east-1:068259748700:log-group:/aws/ec2/lab-1-rds-app",
      "deletionProtectionEnabled": false
    }
  ]
}
```

7.5 Verify DB Failure Logs Appear

Simulate failure (examples):

- Stop RDS

- Change DB password in Secrets Manager without updating DB

- Block SG temporarily

Then check logs:

aws logs filter-log-events \

```
--log-group-name /aws/ec2/lab-1-rds-app \
--filter-pattern "ERROR"
```

```
PS C:\Users\mmauv> aws logs filter-log-events --log-group-name /aws/ec2/lab-1-rds-app --filter-pattern "ERROR"
{
  "events": [
    {
      "logStreamName": "i-01c8c0e651862f3a4/rdsapp-err",
      "timestamp": 1770584133499,
      "message": "[2026-02-08 20:55:33,443] ERROR in app: Exception on /init [GET]",
      "ingestionTime": 1770584138525,
      "eventId": "39485345613722328105213041854798822824213807709323526144"
    },
    {
      "logStreamName": "i-01c8c0e651862f3a4/rdsapp-err",
      "timestamp": 1770584135755,
      "message": "[2026-02-08 20:55:35,695] ERROR in app: Exception on /init [GET]",
      "ingestionTime": 1770584138525,
      "eventId": "39485345664032809273098127662103403247308511266815344645"
    },
    {
      "logStreamName": "i-01c8c0e651862f3a4/rdsapp-err",
      "timestamp": 1770584136256,
      "message": "[2026-02-08 20:55:36,041] ERROR in app: Exception on /init [GET]",
      "ingestionTime": 1770584138525,
      "eventId": "39485345675205482617561969856012798101905340381311533065"
    },
    {
      "logStreamName": "i-01c8c0e651862f3a4/rdsapp-err",
      "timestamp": 1770584136507,
      "message": "[2026-02-08 20:55:36,403] ERROR in app: Exception on /init [GET]",
      "ingestionTime": 1770584138525,
      "eventId": "39485345680802969662393156264538263388340079119312617485"
    },
    {
      "logStreamName": "i-01c8c0e651862f3a4/rdsapp-err",
      "timestamp": 1770584136507,
      "message": "[2026-02-08 20:55:36,403] ERROR in app: Exception on /init [GET]",
      "ingestionTime": 1770584138525,
      "eventId": "39485345680802969662393156264538263388340079119312617485"
    }
  ]
}
```

7.6 Verify CloudWatch Alarm

aws cloudwatch describe-alarms --alarm-name-prefix lab-1-db-connection-failure

```
PS C:\Users\mmauv> aws cloudwatch describe-alarms --alarm-name-prefix lab-1-db-connection-failure
{
  "MetricAlarms": [
    {
      "AlarmName": "lab-1-db-connection-failure",
      "AlarmArn": "arn:aws:cloudwatch:us-east-1:068259748700:alarm:lab-1-db-connection-failure",
      "AlarmDescription": "Triggers when DBConnectionsErrors is greater than or equal to 3 in 5 minutes",
      "AlarmConfigurationUpdatedTimestamp": "2026-02-08T15:38:07.321000-05:00",
      "ActionsEnabled": true,
      "OKActions": [],
      "AlarmActions": [
        "arn:aws:sns:us-east-1:068259748700:lab-1-db-incidents"
      ],
      "InsufficientDataActions": [],
      "StateValue": "ALARM",
      "StateReason": "Threshold Crossed: 1 datapoint [5.0 (08/02/26 21:37:00)] was greater than or equal to the threshold (3.0).",
      "StateReasonData": "{\"version\":\"1.0\",\"queryDate\":\"2026-02-08T21:42:27.422+0000\",\"startDate\":\"2026-02-08T21:37:00.000+0000\",\"statistic\":\"Sum\",\"period\":300,\"recentDatapoints\":[5.0],\"threshold\":3.0,\"evaluatedDatapoints\":[{\"timestamp\":\"2026-02-08T21:37:00.000+0000\",\"sampleCount\":5.0,\"value\":5.0}]}",
      "StateUpdatedTimestamp": "2026-02-08T16:42:27.423000-05:00",
      "MetricName": "DBConnectionErrors",
      "Namespace": "Lab/RDSApp",
      "Statistic": "Sum",
      "Dimensions": [],
      "Period": 300,
      "EvaluationPeriods": 1,
      "Threshold": 3.0,
      "ComparisonOperator": "GreaterThanOrEqualToThreshold",
      "TreatMissingData": "notBreaching",
      "StateTransitionedTimestamp": "2026-02-08T16:42:27.423000-05:00"
    }
  ],
  "CompositeAlarms": []
}
```

7.7 Incident Recovery Verification

After restoring correct credentials or connectivity:

curl http://<EC2_PUBLIC_IP>/list

```
PS C:\Users\mmauv> curl http://3.94.87.137/list

StatusCode      : 200
StatusDescription : OK
Content          : <h3>Notes</h3><ul></ul>
RawContent       : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 23
                  Content-Type: text/html; charset=utf-8
                  Date: Sun, 08 Feb 2026 21:59:35 GMT
                  Server: Werkzeug/3.1.5 Python/3.9.25

                  <h3>Notes</h3><ul></ul>
Forms           : {}
Headers         : [[Connection, close], [Content-Length, 23], [Content-Type, text/html; charset=utf-8], [Date, Sun, 08 Feb 2026 21:59:35 GMT]...]
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 23
```

Reflection Questions: Answer all of these

A) Why might Parameter Store still exist alongside Secrets Manager?

- Many industry environments use both due to the benefits both services provide.
- Parameter store is good for:
 - Non-rotating data
 - Shared application parameters
 - Configuration values
 - Environment configuration
 - End points
 - Feature flags
- Secrets Manager is best for storing sensitive data such as:
 - Credentials (API keys, secrets, passwords)
 - Automatic rotation (used with Lambda)
 - Auditing

B) What breaks first during secret rotation?

- You get an application error:
 - **ERROR** in app: **Exception** on /init [GET]

C) Why should alarms be based on symptoms instead of causes?

- It allows you to diagnose where errors and outages are coming from so you can implement proper remediation strategies using logs and metrics.

D) How does this lab reduce mean time to recovery (MTTR)?

- You are notified of an error immediately in the logs via email, thus allowing you to recover the workloads faster.

E) What would you automate next?

- Secrets and incident response.